Lizhong Hu

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APIs & Databases · Automated Validation · Linux/Docker

Profile

Software Engineering undergraduate in a Sino-American dual-degree program, expected to graduate in 2027. Backend/software engineering experience in FastAPI/Flask services, PostgreSQL/MongoDB storage, API regression checks, and Linux/Docker infrastructure.

Education

Quanzhou University of Information Engineering *2023–2027 (expected)*
Software Engineering (Sino-foreign Cooperation) GPA: 3.10/4.00

Slippery Rock University
Information Systems, BS (in progress; expected 2027) GPA: 3.583/4.000

Sino-American 4+0 dual-degree program conducted in China; approved by the Ministry of Education of China.

Backend & Systems Projects

ETIPlus: Community Governance & WeChat Management Backend Mar–Apr 2025

- Built a Flask backend with PostgreSQL for landlord credit scoring, rankings, and anomaly audits, and MongoDB for WeChat message storage.
- Developed APIs for group moderation, regex nickname checks, and Excel/CSV imports; automated regression checks with Postman.

NoteLLM: Personal Learning QA with Verifiable RAG Ongoing

Undergraduate thesis

- Built a FastAPI backend with async SQLAlchemy, PostgreSQL, and pgvector for cosine-similarity retrieval; implemented page-aware extraction and overlapping chunks for PDF, Markdown, and TXT.
- Validated citations against retrieved chunks on the server and stored source excerpts for verification.
- A single baseline on 7 synthetic documents and 34 fixed questions recorded 100% source-level Recall@5, 97.1% automatic citation-to-source match, and 339 ms mean retrieval latency. These measure retrieval and citation-source matching on that synthetic set, not answer accuracy or general-domain performance.

Code & evaluation report

Privacy-Oriented Personal Network Management System Personal infrastructure project

- Built a Cloudflare Workers service for Clash/V2Ray profile distribution, ISP ASN-based node selection, configuration caching, and CDN speed testing.
- Automated Linux VPS egress configuration in Python, using WARP IPv6 alongside native IPv4; added backups, repeatable deployment, connectivity checks, and routing-conflict regression tests.

DOMjudge Online Judge Deployment & Operations Sep 2024–Present

- Maintain a Linux/Docker-based DOMjudge environment for campus training and programming contests.

On-Site Technical Experience

Sinotech Xinrui (Beijing) Co., Ltd. 7–22 Sep 2026

TOC Operator | Gifu, Japan | 16-day on-site assignment

- Supported football broadcasts at Gifu Nagaragawa Stadium, Japan, for the 20th Asian Games Aichi-Nagoya 2026; host broadcaster: China Media Group-International Media Port (CMG-IMP).
- Coordinated pre-match power, intercom, and primary/backup feed checks; monitored TOC signals. Helped engineers isolate an optical-transmitter fault through receiver replacement and attenuation tests; the feed was restored after replacement and end-to-end verification.

Competitive Programming & Awards

- 2026 National Invitational of CCPC (Fujian), 13th Fujian Collegiate Programming Contest: First to Solve Award (Problem A; problem-level award).
- The 2024 ICPC Asia Kunming Regional Contest: Honorable Medal.
- 15th Lanqiao Cup: First Prize, Fujian Provincial Round, Python Programming (University Group B), 2024.

Technical Skills

Languages & algorithms: C++, Python, Java, JavaScript, Go; algorithms and data structures.
Backend & databases: FastAPI, Flask, RuoYi, PostgreSQL, MongoDB, MySQL; REST API design and integration.
Systems & operations: Linux, Docker, Git, DOMjudge, Cloudflare Workers/CDN/DNS, WARP egress routing.
Retrieval & AI systems: RAG, LLM API integration, prompt design.
Development workflow: Familiar with Codex, Claude Code, and Antigravity CLI; test, static-check, and build validation.